Yes, you read correct. Water cooling blocks for these components exist and there are certain situations where they are actually useful. Let's take a look at these components and the scenarios where they do benefit from liquid cooling.

RAM

RAM DIMMs have come a long way since the DDR1 days. DDR4 RAM now runs at 1.2 Volts as opposed to DDR1 which ran at 2.5 volts. The reduced power requirement is made possible because of the improved manufacturing process i.e. lithography which made it possible to build smaller transistors which don't require higher voltage to run. As as result, the RAM modules don't generate as much heat as they used to.


Blocks for individual DIMMS

However, should you overclock your RAM, then you will generate a lot of heat. For example, take a high-performing DDR4 kit that's capable of running at 3200 MHz and then start bumping the FSB up till you hit 3666 MHz or more, if you're lucky. Your RAM DIMMs will shoot all the way up to the higher 80s (degrees celsius) on moderate workload. This itself is harmful for the longevity of your RAM modules. This is why high-end kits have functional heatsinks while most normal RAM modules tend to have gimmicky designer heatsinks.

In such use cases, mounting a liquid cooling block will allow your RAM DIMMs to operate at high frequencies in a stable manner. Now you must be wondering that RAM DIMMs are situated right next to each other, so won't that pose an issue when you're implementing water cooling on your RAM modules? The answer is yes, if you're using the module shown above. As a remedy there are adjustable blocks like the one shown below.


Alphacool RAM block